

MME8215: Operations Research

Continuous Linear Programming

Continuous Linear Programming (CLP) spans a wide set of OR problems that can be modeled based on the following assumptions:

- Unknown variables are continuous: The result variable and the decision variables can take any real value. Also, all the different coefficients can take any real value.
- Objectives and constraints are linear expressions: The relationships between the variables are expressed as linear expressions.

CLP establishes a rather simplified model of real world problems. Many assumptions are needed to represent real problems as a set of linear expressions. Yet, CLP is a very powerful tool in *Operations Research* (OR) and it can provide valuable insights to decision-making in different fields. The main reason is that, once these assumptions are taken, linear algebra provides valuable tools to find an optimal solution and analyse it.

The main types of problems that will be covered in the CLP exercises belong to two problem types:

- **Production mix:** Production mix problems help us determine the optimal allocation of resources in the production of good or services to either minimise costs or maximise profits.
- **Blending problems:** In blending problems, the objective is to find the optimal combination of components in a set of products that either minimise costs or maximise profits.

1 Setup

In CLP, the objective variable, noted as z , can take any real value:

$$z \in \mathbb{R}.$$

The value of this unknown variable z is a function of another set of unknowns, which are called decision variables and that represent our decisions. These decision variables are noted as x_j , where j is an integer (sub)index that goes from 1 to n :

$$z = f(x_1, x_2, \dots, x_n)$$

and $x_1, x_2, \dots, x_n \in \mathbb{R}$.

Therefore, we will have n different decision variables, that are going to be continuous. And, in extension, z is also going to be continuous. So now we know from where does the C in *CLP* comes from.

Next, in *CLP*, the function f is a linear function. What we want is to find the **optimal value** of z , which, depending on the type of problem will be either the maximum value (e.g. maximise profits) or the minimum value (e.g. minimise costs).

Hence, we want to maximise or minimise z , which as we said is a linear function and therefore can be expressed as the sum of the product of the decision variables multiplied by a set of coefficients, which are noted as c_1 to c_n .

$$\max \text{ or } \min z = f(x_1, x_2, \dots, x_n) = c_1 \cdot x_1 + c_2 \cdot x_2 + \dots + c_n \cdot x_n.$$

Now, we know that linear functions do not have any absolute minimum or maximum, but the objective function is subject to a set of constraints, which are also linear functions of the decision variables, that is, the sum product of the decision variables times a set of coefficients noted as a_{ij} must be less than, greater than, or equal to another coefficient noted as b_j .

s.t.

$$\begin{aligned} a_{11} \cdot x_1 + a_{12} \cdot x_2 \dots + a_{1n} \cdot x_n &\leq b_1 \\ a_{21} \cdot x_1 + a_{22} \cdot x_2 \dots + a_{2n} \cdot x_n &\leq b_2 \\ &\dots \\ a_{m1} \cdot x_1 + a_{m2} \cdot x_2 \dots + a_{mn} \cdot x_n &\leq b_m \end{aligned}$$

Note that coefficients have two sub-indexes, the second is equal to the index of the corresponding decision variable, and the first is equal to the sub-index of b . These new index i ranges from 1 to m and therefore, our optimisation function is subject to a set of m constraints. We will refer to the expressions to the left of the inequality as the Left Hand Side (LHS), and to the other side of the relationship as the Right Hand Side (RHS).

Now, note that we can use the sum operator to represent the objective function in a more compressed form, that is, the LHS is equal to the sum of the product of c_j times x_j for j equal to 1 up to n . Note that this can also be represented as the dot product of two vectors, the vector x with the decision variables and the vector c of the coefficients.

$$\max \text{ or } \min z = c_1 \cdot x_1 + c_2 \cdot x_2 + \dots + c_n \cdot x_n = \sum_{j=1}^n c_j \cdot x_j = c \cdot x$$

where $x = [x_1, x_2, \dots, x_n]^T$ and $c = [c_1, c_2, \dots, c_n]^T$.

We can do the same transformations to the left hand sides of the constraints, expressing them as the sum product of the decision variables times the left hand side coefficients. And, we may as well express all the left hand sides of the constraints as the product of a matrix A that contains all the different coefficients times the decision variable vector.

$$\begin{aligned}
& \max \text{ or } \min \quad z = \sum_{j=1}^n c_j \cdot x_j \\
& \text{subject to} \\
& \quad \sum_{j=1}^n a_{1j} \cdot x_j \leq b_1, \\
& \quad \sum_{j=1}^n a_{2j} \cdot x_j \leq b_2, \\
& \quad \dots \\
& \quad \sum_{j=1}^n a_{mj} \cdot x_j \leq b_m.
\end{aligned}$$

Or, we can write

$$\begin{aligned}
& \max \text{ or } \min \quad z = c \cdot x \\
& \text{subject to} \\
& \quad A \cdot x \leq b.
\end{aligned}$$

where $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & & & \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$ and $b = [b_1, b_2, \dots, b_m]$.

These alternative compact forms allow us to deal with problems with an arbitrary number of decision variables and an arbitrary number of constraints.

2 Types of problems that can be solved with CLP

2.1 Production Mix Problems

In production mix problems, the decision maker needs to create products to satisfy demand using limited resources. The objective is to maximise the profits, taking into account the demand and the limited availability of resources. Production mix problems represent the simplest form of optimization problem in operations research, where the objective is to maximise a utility function (in this case profits) bounded by a set of constraints. Normally, production mix problems are not found in real world applications, but rather, they are part of larger problems.

Objective function Let us note the objective variable profits as z , expressed in a monetary unit (e.g. €).

- z : profits in €.

The decision maker needs to decide how many units of each product type to manufacture. Let us assume there are n different products. Now, let us note the decision variables $x_j \in \mathbb{R}$ as the amount of product $j \in [1, n]$ to be produced, in a production unit:

- x_j : Quantity of product j to be produced in product units.

The objective is to maximize profits. Let us note the profit per unit of product j as p_j :

- p_j : Profit per unit of product j to be produced in €/unit.

Now, the objective function is to maximise the sum of the product of the number of units produced times the profit per unit, for each product type:

$$\max z = x_1 \cdot p_1 + x_2 \cdot p_2 + \dots + x_n \cdot p_n = \sum_{j=1}^n x_j \cdot p_j.$$

Demand constraints There might be requirements on the demand. Let us note the demand of product j as d_j . The demand constraints will be of the type:

$$\begin{aligned} x_1 &\geq d_1, \\ x_n &\geq d_2, \\ &\dots \\ x_n &\geq d_n. \end{aligned}$$

Or $x_j \geq d_j \forall j \in [1, n]$.

Availability constraints Different limited resources are used to produce each product. Let us assume that we have m different resources. Let us note as b_i as the quantity available of resource i , $i \in [1, m]$, expressed in a given unit:

- b_i : Availability of resource i in resource unit.

Further, to produce a unit of a product j we will require a given amount of resource i . Let us note as a_{ij} the requirement of resource i to produce a unit of product j :

- a_{ij} : Requirement of resource i to produce a unit of product j in resource unit / product unit.

Then, for each resource i , we need to ensure that the total amount used is less or equal the total amount available. The amount used to produce a specific number of products will be the number of units products (e.g. x_j) multiplied by the amount required to produce one unit x_{ij} . If we sum across all products, we obtain the total amount used, which needs to be lower than the availability. Let's take the resource indexed 1 as an example:

$$x_1 \cdot a_{11} + x_2 \cdot a_{12} + \dots + x_n \cdot a_{1n} \leq b_1.$$

Note that we can write this down this constraint in a compact form like:

$$\sum_{j=1}^n x_j \cdot a_{ij} \leq b_1.$$

Now, we need to take into account the other resources (we can write all this m equations in a compact form):

$$\sum_{j=1}^n x_j \cdot a_{ij} \leq b_m.$$

That is, for every resource, we need to ensure that the sum of the quantities used to produce each type of product are less or equal than the total amount available.

Summary Putting it all together, the generic formulation of the production mix problem as a CLP is:

$$\begin{aligned} \max \quad & z = \sum_{j=1}^n x_j \cdot p_j \\ \text{subject to} \quad & \sum_{j=1}^n x_j \cdot a_{ij} \leq b_i, \forall i \in [1, m] \\ & x_j \geq d_j, \forall j \in [1, n] \end{aligned}$$

Example 1. *Production Mix:* A company produces two kinds of products, $P1$ and $P2$. The selling price of $P1$ is 300€. The selling price of $P2$ is 250€.

- $P1$ requires 2 operator man-hours and 1 hour of machine operating time.
- $P2$ requires 1 operator man-hours and 3 hours of machine time.
- Operator man-hours cannot be higher than 40 hours per day.
- Machine operating time cannot be higher than 45 hours per day.
- Marketing states that $P1$ units per day cannot be higher than 12.

How many units of $P1$ and $P2$ should the company produce every day to **maximise income**?

Model We want to find the maximum solution to:

$$\max x = 300 \cdot x_1 + 250 \cdot x_2.$$

This is our objective function. The decision variables are:

- x_1 : units of Product 1,
- x_2 : units of Product 2.

The objective function is subject to the following constraints:

$$\begin{aligned} 2 \cdot x_1 + x_2 &\leq 40, \\ x_1 + 3 \cdot x_2 &\leq 45, \\ x_1 &\leq 12. \end{aligned}$$

We define the problem in its standard form as:

$$\max z = 300 \cdot x_1 + 250 \cdot x_2 + 0 \cdot s_1 + 0 \cdot s_2 + 0 \cdot s_3.$$

And the constraints as:

$$\begin{aligned} 2 \cdot x_1 + x_2 + s_1 &= 40, \\ x_1 + 3 \cdot x_2 + s_2 &= 45, \\ x_1 &= 12. \end{aligned}$$

2.2 Blending Problems

In blending problems, the decision maker needs to find the optimal blending (mixing or combination) of materials or resources to create **products** that satisfy a given demand. Each resource or material will have a different costs and therefore, in its primary form, the objective is to minimise the overall costs of the materials needed to satisfy the demand. Additionally, the required materials or resources needed may be subject to availability constraints (i.e. we cannot consider that the amount of materials or resources available is unlimited).

Objective function Let us assume that we have m different materials or resources to be considered in the problem and n different products.

Let us note the cost of material i as $c_i, i \in [1, m]$. Let us also note x_{ij} the amount of resource or material i used to produce products $j, j \in [1, n]$.

Note that these variables x_{ij} represent our decision variables: we are looking for the optimal combination of resources or materials to manufacture our products. Note also that we have $m \cdot n$ different decision variables (the combination of products and materials).

With these definitions, our objective function can be defined as:

$$\min z = \sum_{i=1}^m \sum_{j=1}^n c_i \cdot x_{ij}.$$

That is, the objective is to minimize the overall costs, which is the sum of the cost of the different materials needed to manufacture all our products.

Availability constraints Since the amount of resources or materials available is limited, let us plug in the availability constraints, let us note as a_m as the availability of the material or resource m . The total amount used to manufacture all products cannot exceed the availability of that specific material or resource. For instance, for material or resource 1, the sum of the amount used to manufacture the different products cannot exceed the availability a_1 , that is:

$$\sum_{j=1}^n x_{1j} \leq a_1.$$

The same applies to other material or resources:

$$\sum_{j=1}^n x_{2j} \leq a_2,$$

...

$$\sum_{j=1}^n x_{mj} \leq a_m.$$

We can write this set of constraints in a compact form as:

$$\sum_{j=1}^n x_{ij} \leq a_i, \forall i \in [1, m].$$

Demand constraint As mentioned above, we need to satisfy the demand for every product, which means that the total amount of a product that we need to manufacture needs to be greater or equal than its demand. Let us note d_j as the demand for product j .

Note that we do not have explicitly defined a decision variable for the quantity of each product to manufacture, but in terms of the blending problem, if all the materials are considered, it is equivalent to think of the total amount of a product as the total amount of the materials needed to manufacture a product. Hence, the demand constraints can be expressed as:

$$\begin{aligned} \sum_{j=1}^m x_{i1} &\geq d_1, \\ \sum_{j=1}^m x_{i2} &\geq d_2, \\ &\dots \\ \sum_{j=1}^m x_{im} &\geq d_n, \\ &\dots \end{aligned}$$

Note that we can express this set of constraints in a compact form as:

$$\sum_{i=1}^m x_{ij} \geq d_j, \forall j \in [1, n].$$

Additional requirements: Minimum amount We may find additional requirements in the problem definition, for instance, we may find a requirement for a minimum amount of an ingredient. Sometimes the requirement is expressed as a percentage or ratio of the total composition of the product.

As an example, let us define that the minimum amount of material or resource 2 for product 3 needs to be 20%, or 0.2. If we are dealing with all the materials and resources used, we can again assume that the amount of material or resource x_{23} needs to be at least 20% of the sum of all ingredients used to manufacture product 3, that is:

$$x_{23} \leq 0.2 \sum_{i=1}^m x_{i3}.$$

We could bring all the dependant coefficients to the left-hand side and express this constraint as:

$$\begin{aligned} 0 &\leq -1 \cdot x_{23} + 0.2 \sum_{i=1}^m x_{i3} \\ 0 &\leq -0.8 \cdot x_{23} + 0.2 \sum_{i=1, i \neq 2}^m x_{i3} \\ 0.8 \cdot x_{23} - 0.2 \sum_{i=1, i \neq 2}^m x_{i3} &\geq 0 \end{aligned}$$

Combination of production mix and blending problems Instead of just minimizing the overall cost, our objective can be to maximise profit. This can be regarded as a combination of the production mix and blending problems. Let us define the unitary selling prices of our n products as $[p_1, p_2, \dots, p_n]$.

If we focus on one product, e.g. product 1, the profits can be calculated as the difference between the selling price and the cost obtained for the total amount of the products sold. Again, if we can

assume that the total amount of ingredients is equal to the total amount of products, expressed in the same unit as the unitary selling price, the overall profits for product 1 can be expressed as:

$$p_1 \cdot \sum_{i=1}^n x_{i1} - \sum_{i=1}^n c_i \cdot x_{i1}.$$

Our objective is to maximize the overall profits, so we need to factor in the profits of the rest of the products with:

$$\max z = \sum_{j=1}^n p_j \cdot \sum_{i=1}^n x_{ij} - \sum_{i=1}^n c_i \cdot x_{ij}.$$

Now, we can re-arrange the terms of the summations as:

$$\max z = \sum_{j=1}^n \sum_{i=1}^n (p_j - c_i) \cdot x_{ij}.$$

Example 2. *Blending Problem:* The Kahuna company manufactures sausages using three kinds of meat. The relevant information about the ingredients is provided in the table below:

Ingredient	Cost (€/kg)	Availability (kg)
Pork	4.31	30
Wheat	2.46	20
Starch	1.86	17

The company makes two types of sausages:

- Economy ($\geq 40\%$ Pork)
- Premium ($\geq 60\%$ Pork)

One sausage has 50 grams (0.05 kg). According to government regulations, the most starch we can use in our sausages is 25%. The company has a contract with a butcher, and has already purchased 23 kg pork, that will go bad if it's not used. The company has a demand for 350 economy sausages and 500 premium sausages.

Write a linear program to figure out the most cost-effective recipe to manufacture our sausages.

Example 3. *Petroleum Blending:* Harry Stamper is an engineer working on a petroleum company. The company produces three grades of motor oil – *super*, *premium*, and *extra*. The three grades are made from the same 3 ingredients, named component 1, component 2 and component 3. The company wants to determine the optimal mix of the 3 components in each grade of motor oil that will maximize profit.

The maximum availability of each component and their cost per barrel are as follows: To ensure

Component	Max barrels available / day	Cost / barrel
1	4500	12€
2	2700	10€
3	3500	14€

the appropriate blend, each grade must have a minimum amount of component 1 plus a combination of other components as follows:

Grade	Component 1	Component 2	Component 3	Selling price/barrel
Super	$\geq 50\%$	$\leq 30\%$	-	23€
Premium	$\geq 40\%$	-	$\leq 25\%$	20€
Extra	$\geq 60\%$	$\geq 10\%$	-	18€

The company wants to make at least 3000 barrels of each blend. Formulate an LP to help Harry Stamper find the optimal blend that maximises profit.

Analysis questions (Do this part once you have reviewed CLP analysis including sensitivity analysis questions)

- What is the total profit obtained by the company with the Extra grade with the solution provided by the solver?
- What is the maximum value that the profit per barrel of component 2, in the Extra grade, can take without changing the base of the optimal solution?
- If the cost of the barrel of component 3 is increased by 1€, will the optimal composition change? Motivate your response
- Do we use every barrel available of every component in the optimal solution? Or do we have any surplus availability for any component? Motivate your response
- The provider of Component 2 may have a problem with the delivery and the availability may be reduced to 400 barrels. Will this shortage affect the optimal composition? Motivate your response
- Grace wants to use the results to renegotiate the terms and conditions with the suppliers of the different components. Which component is a better candidate to increase the availability? How would you use the model to negotiate new terms with the provider, assuming you do not want any change in the base of the optimal solution? Motivate your response

Example 4. *Blending craft beer:* Duff Beer is an american brewery that sells alcoholic beverages worldwide. Duff beer has a very large market share in the US. One of the new business lines of Duff Beer is in the craft beer sector. Duff beer buys 3 types of wort from local suppliers. The availability of wort (in liters) and the price for each of them is specified in the table below. The

Wort	Availability (liters)	Cost (€cents/liter)
Type 1	2500	120
Type 2	1200	95
Type 3	2000	150

blend of the three types of wort must comply with the following quality requirements:

- For Beer Angels, no less than 60% of Type 3 and no more of 20% of Type 2. The price of one bottle of Beer Angel is 4.10€/liter.
- For Beer Beast, no less than 15% of Type 3 and no more than 60% of Type 2. The price of one bottle of Beer Beast is 2.80€/liter.
- For Beer Cactus, no more than 50% of type 2. The selling price is 2.45€/liter.

Write a linear program to find the most profitable blend of wort to elaborate the three kinds of beers.

3 Standard Form

Some algorithms like the Simplex need to operate with problems formulated in a standard form. Every CLP problem described in the setup section can be converted to a standard form where:

- The objective function is of type **maximise**.
- All the constraints are of type **equal**.

That is, any CLP problem can be expressed in the standard form as:

$$\begin{aligned} \max \quad & z = \sum_{i=1}^n c_i \cdot x_i = c^T \\ \text{subject to} \quad & a_{11} \cdot x_1 + a_{12} \cdot x_2 + \dots + a_{1n} \cdot x_n + s_1 = b_1 \\ & a_{21} \cdot x_1 + a_{22} \cdot x_2 + \dots + a_{2n} \cdot x_n + s_2 = b_2 \\ & \dots \\ & a_{m1} \cdot x_1 + a_{m2} \cdot x_2 + \dots + a_{mn} \cdot x_n + s_m = b_m \end{aligned}$$

which can be expressed using a matrix product as:

$$A \cdot x = b$$

and $x_i \geq 0, \forall i$.

3.1 Conversion of minimization problems

If we need to convert a problem where the objective function is of type minimize to the standard form, we need just need to make a change of variable:

$$z = -z.$$

With this change of variable the objective function:

$$\min z = c \cdot x$$

is equivalent to:

$$\max z = -c \cdot x.$$

And now the objective function is of type maximise.

3.2 Conversion of constraints and slack variables

Note that we have introduced a new set of decision variables $s = [s_1, s_2, \dots, s_n]$ used to make all the constraints of type equal. These are new variables that are introduced to change the type of constraint to a constraint of type equal. Imagine for instance that our problem has a constraint of type less or equal:

$$a_{11} \cdot x_1 + a_{12} \cdot x_2 + \dots + a_{1n} \cdot x_n \leq b_1.$$

Since the left hand side is less or equal than the right hand side, we can make the left hand side equal to the right hand side by adding a non-negative variable s_1 to the left hand side:

$$a_{11} \cdot x_1 + a_{12} \cdot x_2 + \dots + a_{1n} \cdot x_n + s_1 = b_1.$$

Obviously, we need to add a non-negativity constraint to the new variable to make sure that both constraints are equivalent: $s_1 \geq 0$.

Equivalently, if we have a constraint of type greater or equal, we can make the same modification, but now, the new variable needs to be lower or equal than zero to make the problem equivalent. For instance, let us consider without loss of generality that we have the following constraint in our problem:

$$a_{21} \cdot x_1 + a_{22} \cdot x_2 + \dots + a_{2n} \cdot x_n \geq b_2.$$

This is equivalent to:

$$a_{21} \cdot x_1 + a_{22} \cdot x_2 + \dots + a_{2n} \cdot x_n + s_2 = b_2.$$

where $s_2 \leq 0$.

Also, if our constraint is of type equal, we do not need to introduce any change, but just to make sure we treat all constraints in the same way, let us introduce again a new variable:

$$a_{31} \cdot x_1 + a_{32} \cdot x_2 + \dots + a_{3n} \cdot x_n + s_3 = b_3.$$

This is equivalent to:

$$a_{31} \cdot x_1 + a_{32} \cdot x_2 + \dots + a_{3n} \cdot x_n + s_3 + s_3 = b_3.$$

where $s_3 = 0$.

3.3 Slack variables

These new variables are called **slack variables**, and they represent the difference between the left hand side and the right hand side. When slack variables are non-negative (meaning that the constraint is of type less or equal), they are called **surplus** variables. Note that normally these constraints represent limited resources, so the slack value will represent the surplus of resource that we have available in the optimal solution. When slack variables are non-positive (meaning that the constraint is of type greater or equal), they are called **excess** variables. Note that these types of constraints will normally represent additional requirements, so the value of the slack variable will represent the excess of the solution with respect to the requirement. When slack variables are zero, they are referred to **artificial variables**. Since they must be zero, they need to be treated differently by the solvers.

3.4 Conversion of negative decision variables

Finally, note that in the standard form, all the decision variables need to be non-negative, we can achieve this with a change of variable:

$$\begin{aligned} x &= x^i - x^{ii} \\ x^i, x^{ii} &\in [0, \infty] \end{aligned}$$

That is, we replace the decision variable x by two non-negative decision variables, when x^{ii} is greater than x^i , x takes negative values.

3.5 Summary

In order for the standard form to hold, we need to plug in the new variables into the model. We first introduce the slack variables into our decision variable vector, so that it becomes:

$$x = [x_1, x_2, \dots, x_n, s_1, s_2, \dots, s_p].$$

To make sure that there is no impact in the objective function, we multiply the non-zero slack variables by a factor of 0. Since artificial variables are equal to zero, we multiply them by an arbitrarily large negative number $-M$. This number needs to be introduced by algorithms such as the Simplex. Let us assume without loss of generality that we have p non-zero slack variables and k artificial variables. Let us note artificial variables as $a_1, a_2, \dots, a_k$. Our objective function becomes:

$$\max z = \sum_{i=1}^n c_i \cdot x_i + 0 \cdot s_1 + 0 \cdot s_2 + \dots + 0 \cdot s_p - M \cdot a_1 - M \cdot a_2 - \dots - M \cdot a_k.$$

And the constraints:

$$\begin{aligned} a_{11} \cdot x_1 + a_{12} \cdot x_2 \dots + a_{1n} \cdot x_n + s_1 + 0 \cdot s_2 + \dots + 0 \cdot s_p + 0 \cdot a_1 + 0 \cdot a_2 + \dots + 0 \cdot a_k &\leq b_1 \\ a_{21} \cdot x_1 + a_{22} \cdot x_2 \dots + a_{2n} \cdot x_n + 0 \cdot s_1 + s_2 + \dots + 0 \cdot s_p + 0 \cdot a_1 + 0 \cdot a_2 + \dots + 0 \cdot a_k &\leq b_2 \\ &\dots \\ a_{p1} \cdot x_1 + a_{p2} \cdot x_2 \dots + a_{pn} \cdot x_n + 0 \cdot s_1 + 0 \cdot s_2 + \dots + s_p + 0 \cdot a_1 + 0 \cdot a_2 + \dots + 0 \cdot a_k &\leq b_p \\ a_{(p+1)1} \cdot x_1 + a_{(p+1)2} \cdot x_2 \dots + a_{(p+1)n} \cdot x_n + 0 \cdot s_1 + 0 \cdot s_2 + \dots + 0 \cdot s_p + a_1 + 0 \cdot a_2 + \dots + 0 \cdot a_k &\leq b_{(p+1)} \\ a_{(p+2)1} \cdot x_1 + a_{(p+2)2} \cdot x_2 \dots + a_{(p+2)n} \cdot x_n + 0 \cdot s_1 + 0 \cdot s_2 + \dots + 0 \cdot s_p + 0 \cdot a_1 + a_2 + \dots + 0 \cdot a_k &\leq b_{(p+2)} \\ &\dots \\ a_{m1} \cdot x_1 + a_{m2} \cdot x_2 \dots + a_{mn} \cdot x_n + 0 \cdot s_1 + 0 \cdot s_2 + \dots + 0 \cdot s_p + 0 \cdot a_1 + 0 \cdot a_2 + \dots + a_k &\leq b_m \end{aligned}$$

3.6 Standard tabular representation

We can express the standard form as stated at the beginning of this section as:

$$\begin{aligned} \max \quad & z = c \cdot x \\ \text{subject to} \quad & A \cdot x = b \end{aligned}$$

where

$$\begin{aligned} z &= [x_1, x_2, \dots, x_n, s_1, s_2, \dots, s_p, a_1, a_2, \dots, a_k]^T \\ c &= [c_1, c_2, \dots, c_n, 0, \dots, 0, -M, \dots, -M]^T \\ A &= \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & 1 & 0 & \dots & 0 \\ a_{21} & a_{22} & \dots & a_{2n} & 0 & 1 & \dots & 0 \\ \dots & & & & & & & \\ a_{m1} & a_{m2} & \dots & a_{mn} & 0 & 0 & \dots & 1 \end{bmatrix} \end{aligned}$$

x and c are two $n + m$ vectors and A is a $m \times (n + m)$ matrix where the right most n columns represent the original LHS coefficients and the right most n columns represent the $n \times n$ identity matrix.

Now, if we move all the dependent terms to the LHS, the objective function becomes:

$$z - c_1 \cdot x_1 - c_2 \cdot x_2 - \dots - c_n \cdot x_n - 0 \cdot s_1 - 0 \cdot s_2 - \dots - 0 \cdot s_p + M \cdot a_1 + M \cdot a_2 + \dots + M \cdot a_k = 0$$

z	x_1	x_2	...	x_n	s_1	s_2	...	s_p	a_1	a_2	...	a_k	RHS
1	$-c_1$	$-c_2$	...	$-c_n$	0	0	...	0	-M	-M	...	-M	0
0	a_{11}	a_{12}	...	a_{1n}	1	0	0	...	0	0	...	0	b_1
0	a_{21}	a_{22}	...	a_{2n}	0	1	0	...	0	0	...	0	b_2
...	...	...	...	...	...	...	...	...	...	...	...	...	...
0	a_{m1}	a_{m2}	...	a_{mn}	0	0	0	...	0	0	...	1	b_m

We can write all the coefficients in a data frame where each column represents the coefficients of a variable and each row represents an equation:

This representation of the problem is called the standard tabular form.

4 Simplex example

We will use a problem as an example to show the simplex method.

Problem Definition Production Mix A company produces two kinds of products, P1 and P2. The selling price of P1 is 300€ The selling price of P2 is 250€

- P1 requires 2 operator man hours and 1 hour of machine operating time.
- P2 requires 1 operator man hours and 3 hours of machine time.
- Operator man hours cannot be higher than 40 hours per day.
- Machine operating time cannot be higher than 45 hours per day.
- Marketing states that P1 units per day cannot be higher than 12.

How many units of P1 and P2 should the company produce every day to maximise income?

Model We want to find the maximum solution to:

$$\max z = 300 \cdot x_1 + 250 \cdot x_2.$$

This is our objective function. The decision variables are:

- x_1 : units of Product 1,
- x_2 : units of Product 2.

The objective function is subject to the following constraints:

$$\begin{aligned} 2 \cdot x_1 + x_2 &\leq 40, \\ x_1 + 3 \cdot x_2 &\leq 45, \\ x_1 &\leq 12. \end{aligned}$$

We define the problem in its standard form as:

$$\max z = z = 300 \cdot x_1 + 250 \cdot x_2 + 0 \cdot s_1 + 0 \cdot s_2 + 0 \cdot s_3.$$

Where:

- s_1 : Slack variable of the operator man hour constraint ($s_1 \geq 0$),
- s_2 : Slack variable of the machining time constraint ($s_2 \geq 0$),
- s_3 : Slack variable of the operator man hour constraint ($s_3 \geq 0$).

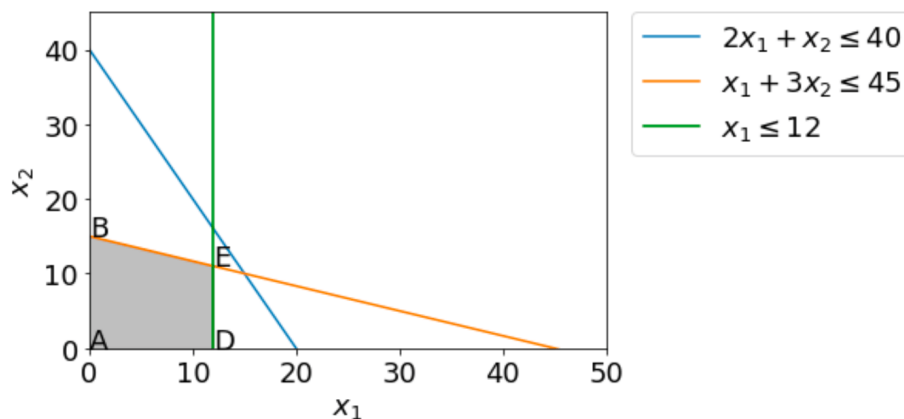
And thus, by entering the slack variables, we can now express the constraints as:

$$2 \cdot x_1 + x_2 + s_1 = 40,$$

$$x_1 + 3 \cdot x_2 + s_2 = 45,$$

$$x_1 + s_3 = 12.$$

Solution We will use the Simplex Method and iterate between the graphical and the algebraic form of the problem to describe the results. First let us represent the problem: The feasible region



(in grey) contains all possible solutions to the objective function. Let us now represent the same problem in a basic Tableau: From the Tableau, we can easily find a first Feasible Basic Solution

Z	x_1	x_2	s_1	s_2	s_3	RHS
1	-300	-250	0	0	0	0
0	2	1	1	0	0	40
0	1	3	0	1	0	45
0	1	0	0	0	1	12

(FSB), by making x_1 and x_2 non basic ($= 0$). The three equations of the constraints provide us the value of the slack variables in this solution:

$$x_1 = 0, x_2 = 0, s_1 = 40, s_2 = 45, s_3 = 12.$$

Let us note this solution as $(0, 0, 40, 45, 12)$ and $z = 0$. Note that this FSB corresponds to Vertex A in the above figure. *How can we know if it is optimal?* We have 2 non basic variables x_1 and x_2 which have negative coefficients in the objective function and positive values of these variables would increase z . If we increase x_1 in one unit, z would increase 300 units and if we increase x_2 in one unit z would increase 250. Therefore, the solution is not optimal, since we could find a higher z for a positive value of these variables in the objective function.

Hence, this is not the optimal solution and we need to look for it in another vertex. From point A , we could either move to vertex B or to vertex D . Moving towards D would mean making x_1 higher than zero, thus making it a **basic variable**. Likewise, moving towards B would mean making x_2 higher than zero and basic (enter the base). We select x_1 as the **entering basic variable**, since its coefficient (300) is higher and therefore z will increase at a higher rate when we move towards D . We have just selected the pivot variable and the pivot column in the Tableau.

Now, we are going to move towards D , which is equivalent to make $x_1 = \delta$. As this value increases, the objective value will also increase, but we have to bear in mind that all variables must be positive and therefore δ cannot take any given value: we have to make sure the solution is **feasible**. If we write down all the constraints in algebraic form, we can find the possible values of δ

$$\begin{aligned} 2 \cdot \delta + 1 \cdot 0 + s_1 &= 40, \\ 1 \cdot \delta + 3 \cdot 0 + s_2 &= 45, \\ 1 \cdot \delta + s_3 &= 12. \end{aligned}$$

To ensure that all the surplus variables are positive δ cannot exceed the maximum positive value that guarantees that all the surplus variables are non negative. The strategy to find a feasible solution is to make one slack variable **non-basic** (equal to zero), but making sure that all the constraints are met. We only have δ and a slack variable in each expression, this means that the maximum value it can take is $\delta = \min\{40/2, 45/1, 12/1\}$. Any value above this makes the solution unfeasible. This is called the **Ratio test**. If we make $x_1 = \delta = 12$ then that implies that s_3 is now a non-basic variable. With the Ratio test, we have selected s_3 as the leaving variable which will no longer be basic in the FSB. Let us add two new columns to the basic Tableau, one to the right to indicate which variable is the basic variable in the equation and another one to the left to express this Ratio test.

Basic	z	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
.	1	-300	-250	0	0	0	0	.
s_1	0	2	1	1	0	0	40	40/2=20
s_2	0	1	3	0	1	0	45	45/1=45
s_3	0	1	0	0	0	1	12	12/1=12

Now, notice that from the Ratio test, we have easily determined that the variable that leaves the basic solution is s_3 and using the last equation we can easily determine that $x_1 = 12$ in the new solution. We will move to another solution $(12, 0, ?, ?, 0)$ which corresponds exactly to point D . Now notice that in the first iteration, the problem was expressed in a very convenient way since we had only a non-zero (1) value for each basic variable in every row. To accomplish this, we are going to transform the different rows. Since the equations are linearly independent, we can make linear operations in the different equations and still obtain a linear independent system of equations. We want to obtain a system like this:

Basic	z	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
.	1	0	?	0	0	?	?	.
s_1	0	0	?	1	0	?	?	?
s_2	0	0	?	0	1	?	?	?
x_1	0	1	?	0	0	?	?	?

Note that when the problem is in this form, we will have each basic variable in a row with coefficient 1 and the rest with coefficient 0 and we will be able to obtain the value for each basic variable rather easily. First we transform the **leaving** equation, by dividing the entire row by the coefficient of the entering variable in that equation (in this case is 1) and the row remains unaltered:

Basic	z	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
•	1	0	?	0	0	?	?	•
s_1	0	0	?	1	0	?	?	?
s_2	0	0	?	0	1	?	?	?
x_1	0	1/1=1	0/1=0	0/1=0	0/1=0	1/1=1	12/1=12	?

Now, for the rest of the row, we need to make a linear operation that gives a zero in the cell corresponding to the entering variable. To accomplish this, we subtract the **entering equation** multiplied by the value of the coefficient that multiplies the entering variable x_1 .

Basic	z	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
•	1	$-300 - (-300 \cdot 1) = 0$	$-250 - (-300 \cdot 0) = -250$	$0 - (-300 \cdot 0) = 0$	$0 - (-300 \cdot 0) = 0$	$0 - (-300 \cdot 1) = 300$	$0 - (-300 \cdot 12) = 3600$	•
s_1	0	$2 - 2 \cdot 1 = 0$	$1 - 2 \cdot 0 = 1$	$1 - 2 \cdot 0 = 1$	$0 - 2 \cdot 0 = 0$	$0 - 2 \cdot 1 = -2$	$40 - 12 \cdot 2 = 16$	?
s_2	0	$1 - 1 \cdot 1 = 0$	$3 - 1 \cdot 0 = 3$	$0 - 1 \cdot 0 = 0$	$1 - 1 \cdot 0 = 1$	$0 - 1 \cdot 1 = -1$	$45 - 12 \cdot 1 = 33$	?
x_1	0	1	0	0	0	1	12	?

We end up with the following Tableau:

Basic	z	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
•	1	0	-250	0	0	300	3600	•
s_1	0	0	1	1	0	-2	16	?
s_2	0	0	3	0	1	-1	33	?
x_1	0	1	0	0	0	1	12	?

And the new FSB solution is (12, 0, 16, 32, 0) which as noted corresponds to point D .

At this point, z has increased from 0 to 3600, but we still have a coefficient in x_2 that is lower than zero and therefore we could find a higher value if again we move from D to E by increasing x_2 , just as we did with x_1 . We have selected the next entering variable and the pivot column and we can perform the Ratio test again to select the variable leaving the basic solution:

Basic	z	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
•	1	0	-250	0	0	300	3600	•
s_1	0	0	1	1	0	-2	16	16
s_2	0	0	3	0	1	-1	33	11
x_1	0	1	0	0	0	1	12	12

The minimum value that makes all variables positive is 11. Therefore, the variable leaving the solution is s_2 . We first transform the new entering row:

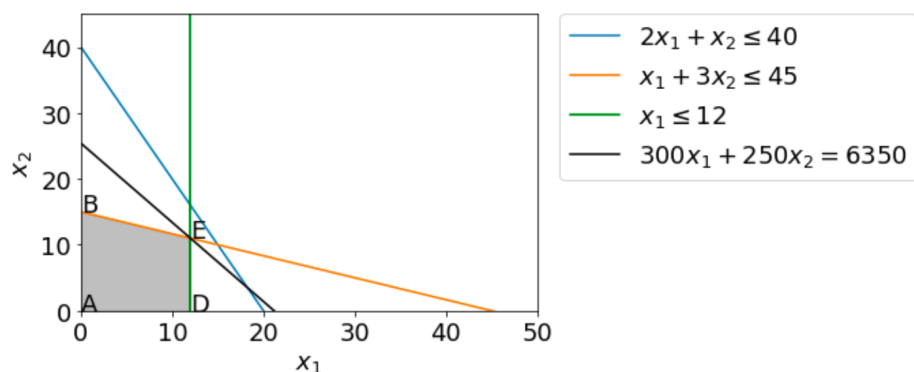
Basic	z	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
.	1	?	0	0	?	?	?	.
s_1	0	0	0	1	?	?	?	?
x_2	0	0	1	0	1/3	-1/3	11	?
x_1	0	1	0	0	?	?	?	?

And then the rest of the rows using linear transformations as we did before we obtain. The next two tables show this. (The second table shows only the variables and RHS)

Basic	z	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
.	1	$0 - (-250 \cdot 0) =$	$-250 - (-250 \cdot 0) =$	$0 - (-250 \cdot 0) =$	$0 - (-250 \cdot 1/3) =$	$300 - (-250 \cdot -1/3) =$	$3600 - (-250 \cdot 11) =$	.
s_1	0	0	$1 - 1 = 0$	$1 - 0 = 0$	$0 - 1/3 = -1/3$	$-2 - (-1/3) = -5/3$	$16 - 11 = 5$	.
x_2	0	0	1	0	1/3	-1/3	11	.
x_1	0	1	0	0	0	1	12	.

z	x_1	x_2	s_1	s_2	s_3	RHS
1	$0 - (-250 \cdot 0) = 0$	$-250 - (-250 \cdot 1) = 0$	$0 - (-250 \cdot 0) = 0$	$0 - (-250 \cdot 1/3) = 250/3$	$300 - (-250 \cdot -1/3) = 650/3$	$3600 - (-250 \cdot 11) = 6350$
0	0	$1 - 1 = 0$	$1 - 0 = 0$	$0 - 1/3 = -1/3$	$-2 - (-1/3) = -5/3$	$16 - 11 = 5$
0	0	1	0	1/3	-1/3	11
0	1	0	0	0	1	12

Now we obtain a new solution $(12, 11, 5, 0, 0)$ where $z = 6350$ and we know for sure that this is a maximum because there are no negative coefficients for any of the non-basic variables that could yield to a higher value of z .



5 Duality

5.1 Introduction

The duality principle states that every optimization problem can be represented and analysed in two different formulations, the **primal** and the **dual**. This means that every problem that we have analysed so far has a **dual** problem formulation with the following characteristics:

- The dual has an optimal solution **if and only if** the primal has an optimal solution (and vice versa).
- The **optimal value** of the objective function both cases (respective optimal solutions) **is the same** (strong duality principle).
- The dual of dual is the primal.
- The solution of the dual problem can be obtained from the solution of the primal and vice versa (using algorithms like the like Simplex)

Basic rules for obtaining the dual The dual of a problem can be obtained applying the following rules:

- If the **primal problem objective function** is of type **maximise**, the **dual problem objective function** is of type **minimise** (and vice versa).
- If the primal has m constraints, the dual has m decision variables, noted as $u_1, u_2, \dots, u_m$.
- The coefficients of the objective function of the dual are the independent terms (RHS) of the constraints of the primal (and vice versa).
- The coefficients of the dependent terms (LHS) of the constraint of the dual can be represented by the transposed matrix A^T that contains the LHS coefficients of the primal (and vice versa).
- All the constraints are of type greater or equal ($\geq$).
- For maximization problems:
 - If the i^{th} constraint of the primal is of type less or equal ($\leq$), then u_i is non-negative (≥ 0).
 - If the i^{th} constraint of the primal is of type greater or equal ($\geq$), then u_i is non-positive (≤ 0).
- For minimization problems:
 - If the i^{th} constraint of the primal is of type less or equal ($\leq$), then u_i is non-positive (≤ 0).
 - If the i^{th} constraint of the primal is of type greater or equal ($\geq$), then u_i is non-negative (≥ 0).
- If any case, if the i^{th} constraint of the primal is of type equal ($=$), then u_i is unrestricted (can be positive or negative).

Summary These rules are summarised in the following table:

Primal problem	Dual problem
Objective of type maximize	Objective of type minimize
Objective of type minimize	Objective of type maximize
Constraint limits, RHS	Decision Variable coefficients
Decision Variable coefficients	Constraint limits, RHS
Constraint coefficient of decision variable i (a_{ij})	Constraint coefficient of constraint i (a_{*ji})

5.2 Simplex example

Let us consider the following generic problem with two decision variables:

$$\begin{aligned}
 \max \quad & z = c_1 \cdot x_1 + c_2 \cdot x_2 \\
 \text{subject to} \quad & \\
 & a_{11} \cdot x_1 + a_{12} \cdot x_2 \leq b_1 \\
 & a_{21} \cdot x_1 + a_{22} \cdot x_2 \leq b_2 \\
 & a_{31} \cdot x_1 + a_{32} \cdot x_2 \leq b_2
 \end{aligned}$$

Applying the rules above, the objective function of the dual is of type minimise, there are 3 decision variables, as the primal has 3 constraints, and the coefficients of the objective function of the dual are the RHS of the primal, hence, the objective function of the dual becomes:

$$\min z = b_1 \cdot u_1 + b_2 \cdot u_2 + b_3 \cdot u_3.$$

Now, we have two constraints, one per every decision variable. The RHS are the coefficients of the objective function, and the dependent coefficient are obtained by transposing the coefficients of the primal, hence the dual problem is subject to the following constraints:

$$\begin{aligned}
 a_{11} \cdot u_1 + a_{21} \cdot u_2 + a_{31} \cdot u_3 &\geq c_1 \\
 a_{12} \cdot u_1 + a_{22} \cdot u_2 + a_{32} \cdot u_3 &\geq c_2
 \end{aligned}$$

Now, since the constraints of the primal are of type $\leq$, the three decision variables are non-negative: $u_1, u_2, u_3 \geq 0$.

5.3 Bounding explanation

One way to deduct the dual is by acknowledging that, if the objective function of the primal is of type maximize and all the constraints are of type less or equal, linear combinations of the constraints provide **upper bounds** for the objective function. For instance, let us consider the following problem:

$$\begin{aligned}
 \max \quad & z = 4 \cdot x_1 + x_2 + 5 \cdot x_3 + 3 \cdot x_4 \\
 \text{subject to} \quad & \\
 & x_1 - x_2 - x_3 + 4 \cdot x_4 \leq 1 \\
 & 5 \cdot x_1 + x_2 + 3 \cdot x_3 + 8 \cdot x_4 \leq 55 \\
 & -x_1 + 2 \cdot x_2 + 3 \cdot x_3 + 5 \cdot x_4 \leq 3
 \end{aligned}$$

Note that, if we multiply the second constraint by a factor of 2, we obtain an equivalent constraint:

$$10 \cdot x_1 + 2 \cdot x_2 + 6 \cdot x_3 + 16 \cdot x_4 \leq 110.$$

Note also that all the LHS coefficients of this constraint are higher or equal than the coefficients of the objective function, and therefore:

$$\max z = 4 \cdot x_1 + x_2 + 5 \cdot x_3 + 3 \cdot x_4 \leq 10 \cdot x_1 + 2 \cdot x_2 + 6 \cdot x_3 + 16 \cdot x_4 \leq 110.$$

This constraint is an upper bound for the objective function, we do not know yet which is the maximum value of z , but we know that it cannot be higher than 110.

Note that we can find another upper bound using a linear combination of the second of third constraint, just by adding both constraints, we obtain:

$$4 \cdot x_1 + 3 \cdot x_2 + 6 \cdot x_3 + 14 \cdot x_4 \leq 58.$$

Again, this linear combination is an upper bound of the objective function, since all the coefficients are higher or equal the coefficients of the objective value, for any value of the decision variables, we have:

$$\max z = 4 \cdot x_1 + x_2 + 5 \cdot x_3 + 3 \cdot x_4 \leq 4 \cdot x_1 + 3 \cdot x_2 + 6 \cdot x_3 + 14 \cdot x_4 \leq 58.$$

Therefore, even if we do not know which is the optimal value of the objective function, 58 is an upper bound, the optimal value cannot exceed this bound.

Note that finding the maximum value of z , is equivalent to finding the **minimum upper bound**, hence, the minimum linear combination of the constraints, or equivalently, the minimum linear combination of the RHSs:

$$\min z = 1 \cdot u_1 + 55 \cdot u_2 + 4 \cdot u_3$$

which yields coefficients that are higher or equal the coefficients of the objective function for every decision variable:

$$\begin{aligned} u_1 + 5 \cdot u_2 - u_3 &\geq 4 \\ -u_1 + u_2 + 2 \cdot u_3 &\geq 1 \\ -u_1 + 3 \cdot u_2 + 3 \cdot u_3 &\geq 5 \\ 4 \cdot u_1 + 8 \cdot u_2 + 5 \cdot u_3 &\geq 3 \end{aligned}$$

Note that this is the dual of the primal problem.

5.4 Complementary slackness

Following the reasoning of the previous section, note that the decision variables of the dual are equivalent to the **Lagrangian multipliers** of the constraints in the Lagrangian of the primal:

$$\max z = \sum_{i=1}^n c_i \cdot x_i + u_1(a_{11} \cdot x_1 + a_{12} \cdot x_2 + \dots + a_{1n} \cdot x_n) + u_2(a_{21} \cdot x_1 \dots)$$

Maximizing the objective function is equivalent to minimizing the part of the Lagrangian multipliers.

But most importantly, Dual variables represent changes in the primal objective function per unit of constraint, that is, if an RHS of a constraint i is modified in 1 unit, the objective function is going to change in u_i units.

For this reason, the decision variables of the dual are referred to as the **shadow prices** of the constraints. The term shadow highlights that they are not factored in the objective function, and the term price means that they represent the impact in the objective function.

It follows that if a constraint is non-binding, meaning that the corresponding slack variable is non-zero, then the corresponding decision variable, or shadow price must necessarily be zero (that is, the shadow price must be zero, because we have a surplus or excess, and a change in the RHS of the constraint is not going to change the objective variable, just the value of the surplus or excess). Conversely, if a constraint is binding (meaning that the corresponding slack variable is zero), then the corresponding dual variable (shadow price) is basic (non-zero), because a change in the RHS of the constraint can cause a change in the objective variable.

Same applies for the dual: If a constraint of the dual is binding (meaning that the slack variable is zero, or non-basic), then the corresponding primal variable is basic. Also, if a constraint of the dual is non-binding (meaning that the slack variable is non-zero, or basic), then the corresponding primal variable is non-basic.

This principle is called **complementary slackness** meaning that the slack values or slackness of dual and primal are complementary.

6 Sensitivity: Algebraic explanation

6.1 Changes in the coefficients of the objective function

Let us start with the Production Mix Problem example, which in the standard form is:

$$\begin{aligned} \max \quad & z = 300 \cdot x_1 + 250 \cdot x_2 + 0 \cdot s_1 + 0 \cdot s_2 + 0 \cdot s_3 \\ \text{subject to} \quad & 2 \cdot x_1 + x_2 + s_1 = 40, \\ & x_1 + 3 \cdot x_2 + s_2 = 45, \\ & x_1 = 12. \end{aligned}$$

The final Tableau is:

Basic	z	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
•	1	0	0	0	250/3	650/3	6350	•
s_1	0	0	0	1	-1/3	-5/3	5	•
x_2	0	0	1	0	1/3	-1/3	11	•
x_1	0	1	0	0	0	1	12	•

Let us assume we want to make a change in coefficient c_1 of magnitude Δc_1 . The objective function becomes:

$$\max z = (300 + \Delta c_1) \cdot x_1 + 250 \cdot x_2.$$

The final Tableau now becomes as the one presented by table 1.

The Simplex method requires that the Tableau is in canonical form. Thus, we need to add the last row times Δc_1 to the first row for this, table 2 shows this.

And taking into account that the coefficient of s_3 needs to be positive for x_1 to remain in the basis, Δc_1 cannot be lower than $-650/3$ and can increase up to infinity.

Let us now assume we want to make a change in coefficient c_2 of magnitude Δc_2 . The objective function becomes:

$$\max z = 300 \cdot x_1 + (250 + \Delta c_2) \cdot x_2.$$

The final Tableau now becomes as the one presented by table 3.

Again, the Simplex method requires that the Tableau is in canonical form. Thus, we need to add the second last row times Δc_2 to the first row for this, table 4 shows this.

And taking into account that the coefficients of s_2 and s_3 need to be positive for x_2 to remain in the basis, Δc_2 cannot be lower than -250 nor higher than 650 .

Table 1

Basic	z	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
•	1	$-\Delta c_1$	0	0	250/3	650/3	6350	•
s_1	0	0	0	1	-1/3	-5/3	5	•
x_2	0	0	1	0	1/3	-1/3	11	•
x_1	0	1	0	0	0	1	12	•

Table 2

Basic	z	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
•	1	0	0	0	250/3	$650/3 + \Delta c_1$	$6350 + 12 \cdot \Delta c_1$	•
s_1	0	0	0	1	-1/3	-5/3	5	•
x_2	0	0	1	0	1/3	-1/3	11	•
x_1	0	1	0	0	0	1	12	•

Table 3

Basic	z	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
•	1	0	$-\Delta c_2$	0	250/3	650/3	6350	•
s_1	0	0	0	1	-1/3	-5/3	5	•
x_2	0	0	1	0	1/3	-1/3	11	•
x_1	0	1	0	0	0	1	12	•

6.2 Changes in the constraint independent terms

Now let us see what is the effect of changing b , for instance decreasing b_1 an amount equal to Δb_1 . In the original problem formulation, this means:

$$2 \cdot x_1 + x_2 + s_1 = 40 - \Delta b_1,$$

$$x_1 + 3 \cdot x_2 + s_2 = 45,$$

$$x_1 + s_3 = 12.$$

Changing the Right Hand Side by Δb_1 is equivalent to changing s_1 an amount of Δb_1 . Note that

Table 4

Basic	z	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
•	1	0	0	0	$250/3 + \Delta c_2/3$	$650/3 - \Delta c_2/3$	$6350 + 11 \cdot \Delta c_2$	•
s_1	0	0	0	1	$-1/3$	$-5/3$	5	•
x_2	0	0	1	0	$1/3$	$-1/3$	11	•
x_1	0	1	0	0	0	1	12	•

s_1 is in the basis, thus for it to remain in the basis, it must satisfy:

$$s_1 = 5 + \Delta b_1 \geq 0,$$

$$\Delta b_1 \geq -5.$$

Now, take the second constraint and let us apply the same change:

$$2 \cdot x_1 + x_2 + s_1 = 40,$$

$$x_1 + 3 \cdot x_2 + s_2 = 45 - \Delta b_2,$$

$$x_1 + s_3 = 12.$$

Again, this is equivalent to:

$$2 \cdot x_1 + x_2 + s_1 = 40,$$

$$x_1 + 3 \cdot x_2 + s_2 + \Delta b_2 = 45,$$

$$x_1 + s_3 = 12.$$

Which can be regarded as a change of Δb_2 in s_2 . Now, if we change s_2 from $s_2 = 0$ to $s_2 = \Delta b_2$ in the final Tableau:

Basic	z	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
•	1	0	0	0	$250/3 \cdot \Delta b_2$	$650/3$	6350	•
s_1	0	0	0	1	$-1/3 \cdot \Delta b_2$	$-5/3$	5	•
x_2	0	0	1	0	$1/3 \cdot \Delta b_2$	$-1/3$	11	•
x_1	0	1	0	0	$0 \cdot \Delta b_2$	1	12	•

The basis remains unchanged as long as all the constraints are still met:

$$s_1 - 1/3 \cdot \Delta b_2 = 5,$$

$$x_2 + 1/3 \cdot \Delta b_2 = 11,$$

$$x_1 + 0 \cdot \Delta b_2 = 12,$$

$$s_1 = 5 + 1/3 \cdot \Delta b_2 \geq 0 (\Delta b_2 \geq -15),$$

$$x_2 = 11 - 1/3 \cdot \Delta b_2 \geq 0 (\Delta b_2 \leq 33),$$

$$x_1 = 12 - 0 \cdot \Delta b_2 \geq 0 (\Delta b_2 \leq \infty).$$